

Full-Dimensional Jukes–Cantor Ambiguity in Weakly Tree-Child Level-2 Networks

An exact four-leaf construction and an all-taxa extension

Alec Kriebel
Independent Researcher

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Abstract

We give two explicit four-leaf binary level-2 semi-directed phylogenetic networks whose labelled mixed graphs are nonisomorphic and cannot be related by redirecting their unique triangle, but whose open Jukes–Cantor model images share a regular relatively open subset of full dimension eight. Each network has a tree-child rooting, but neither is strongly tree-child in the semi-directed sense that every admissible rooting is tree-child. The proof is exact: on the physical locus, both models lie on the same irreducible eight-dimensional component cut out by six Fourier equations; an algebraic common point lies strictly inside the positive-definite stochastic parameter space; and rank-eight Jacobian minors are nonzero on both parameterizations. An independent verifier checks all 256 Fourier coordinates in a quadratic number field. Replacing corresponding leaves by identical rooted binary trees has a positive analytic inverse. Consequently, for every $n \geq 4$ there is such a non-triangle ambiguity on n leaves, with a common regular open region of dimension $2n$. The result supplies an exact negative instance among same-type four-leaf level-2 comparisons and gives a precise obstruction to extending identifiability statements from strongly to merely weakly tree-child semi-directed networks.

Keywords. phylogenetic networks; identifiability; Jukes–Cantor model; level-2 networks; tree-child networks; algebraic statistics.

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1 Introduction

A phylogenetic-network topology is scientifically recoverable from exact site-pattern data only when distinct topologies cannot generate the same generic distribution. For trees this question has a long algebraic history; for networks, mixtures over displayed trees introduce additional nonidentifiability. Triangle orientation is one familiar source: under group-based models, redirecting the reticulation inside a small triangle can preserve a full-dimensional part of the model.

This paper exhibits a different phenomenon at level two. The two networks have the same unlabelled theta core, but exchange two labelled pendant components between a vertex on the unique triangle and a vertex off that triangle. Their standard semi-directed graphs are therefore not related by triangle redirection. Nevertheless, their Jukes–Cantor (JC) images meet in full dimension on the open stochastic parameter space.

The tree-child convention is essential. Following the semi-directed terminology of Holtgreffe et al. [6], a semi-directed topology is *weakly tree-child* if it has at least one tree-child rooting and *strongly tree-child* if every admissible rooting is tree-child. The pair below is weakly but not strongly tree-child. It therefore does not contradict the triangle-free strongly tree-child identifiability theorem of Englander et al. [4]; instead it shows that allowing all weakly tree-child topologies introduces a genuinely non-triangle ambiguity.

Ardiyansyah [1] classified several four-leaf simple level-2 comparisons algebraically and identified same-type comparisons that were not resolved by those calculations. The construction below supplies an exact negative instance for one labelled pair of the triangle-containing type. It is distinct from the three-leaf semialgebraic triangle-orientation overlap studied by Currie et al. [3], and from the stacked- reticulation and two-blob local modifications studied by Sullivant [8]. Brits et al. [2] prove full identifiability for level-1 networks modulo triangle redirection and analyze tree-network separation in the presence of specified two-sub-blob phenomena; the construction here is a level-2 comparison and uses no two-sub-blob suppression.

The main result is deliberately local in scope.

Theorem 1.1 (All-taxa full-dimensional ambiguity). *For every $n \geq 4$, there are two leaf-labelled binary standard semi-directed level-2 networks $N_n, N'_n \in W_{TC} \setminus S_{TC}$ such that:*

- (i) N_n and N'_n are nonisomorphic and are not related by ordinary triangle redirection $\mathbb{T}$;
- (ii) their open JC model images have dimension $2n$; and
- (iii) their images share a regular relatively open subset of dimension $2n$.

In particular, the nonidentifiable parameter set contains a nonempty Euclidean-open subset of the full parameter space and cannot be confined to a proper algebraic exceptional set.

The proof starts from an exact four-leaf certificate and then uses a leaf-substitution map with an analytic inverse. Equality of complex model closures alone is never used as evidence of stochastic overlap.

2 Networks and the JC model

2.1 Rooted and semi-directed conventions

Let X be a finite set of taxon labels. A *rooted binary phylogenetic network* on X is a finite directed acyclic graph without parallel arcs in which the root has bidegree $(0, 2)$, a tree vertex has bidegree $(1, 2)$, a reticulation has bidegree $(2, 1)$, and each leaf has bidegree $(1, 0)$ and a unique label in X . We additionally require the root to be the lowest stable ancestor of X : no proper descendant of the root is a stable ancestor of every labelled leaf. A rooted network is *tree-child* if every internal vertex has a child that is a tree vertex or a leaf.

We use the following narrow, reticulation-preserving semi-directed reduction, denoted sd_0 . Mark every arc whose head is a reticulation and undirect every unmarked arc. Delete the root and replace its two incident edges by one edge joining its two children, retaining at either endpoint exactly the arrowhead that was present there before deletion. A rooted presentation is admitted only if this operation produces a simple binary mixed graph: it creates neither a loop nor a parallel edge, and every reticulation vertex and every retained incoming arrowhead survives. No further parallel identification or degree-two cleanup is part of sd_0 . An isomorphism preserves labels, undirected edges, and retained arrowheads. An *admissible rooting* of a semi-directed network is an LSA-valid

rooted binary network whose sd_0 reduction is exactly the given mixed graph. Thus a root site may be an undirected edge or a compatible edge entering a reticulation: inserting the root on $u \rightarrow r$ replaces that edge by $\rho \rightarrow u$ and $\rho \rightarrow r$, and suppressing ρ recovers $u \rightarrow r$. This is the simple, root-suppression convention of Definitions 2.1–2.2 in Englander et al. [4], written here as an explicit map. In particular, it is not the broader cleanup operation sometimes used after taking induced subnetworks: such a cleanup may merge parallel edges or erase reticulation vertices and is not used to define the rooting universe here.

We distinguish

$$\begin{aligned} R_{\text{TC}} &= \{\text{rooted tree-child presentations}\}, \\ W_{\text{TC}} &= \{\text{semi-directed networks with at least one tree-child rooting}\}, \\ S_{\text{TC}} &= \{\text{semi-directed networks for which every rooting is tree-child}\}. \end{aligned}$$

Thus $S_{\text{TC}} \subseteq W_{\text{TC}}$, while R_{TC} is a property of a chosen rooted presentation.

A cut edge is an edge whose deletion disconnects the underlying graph. A *blob* is a maximal connected subgraph containing no cut edge. (For the simple binary mixed graphs used here, this agrees with the nontrivial biconnected-block convention.) A network is *level two* if each blob contains at most two reticulations.

Definition 2.1 (Ordinary triangle redirection). *Suppose a semi-directed network has a triangle Δ . An ordinary triangle redirection T changes the reticulation vertex and the two incoming arrowheads within Δ while leaving the labelled underlying graph and every arrowhead outside Δ unchanged.*

In particular, T cannot move a labelled pendant component from one vertex of the underlying graph to another.

2.2 Fourier parameterization

Identify the four states with $G = \mathbb{Z}_2 \times \mathbb{Z}_2$. Every edge e has one nontrivial Fourier multiplier $x_e \in (0, 1)$; equivalently, its JC transition matrix has entries

$$T_e(a, b) = \begin{cases} (1 + 3x_e)/4, & a = b, \\ (1 - x_e)/4, & a \neq b. \end{cases}$$

Every reticulation r has an inheritance probability $\lambda_r \in (0, 1)$. The root state is uniform on G . Independently at each reticulation, retain one incoming parent with probability λ_r and the other with probability $1 - \lambda_r$; conditional on those choices, states evolve down the resulting displayed tree. This is the open positive-definite parameter space Θ_0 ; no multiplier is zero or one, and no inheritance probability is zero or one.

For a displayed tree T and a character assignment $g = (g_i)_{i \in X} \in G^X$, Fourier diagonalization of the JC transition matrices gives

$$\hat{p}_T(g) = \begin{cases} \prod_{e \in E(T)} x_e^{\mathbf{1}[\oplus_{i \in A_e} g_i \neq 0]}, & \oplus_{i \in X} g_i = 0, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $A_e \mid A_e^c$ is the split induced by e . A network coordinate is the inheritance-weighted sum of (1) over all parent selections. This follows directly from the character calculation for group-based models [5, 7].

Let $\mathcal{M}_N^{\text{JC}}$ be the image of Θ_0 in pattern-probability space and $\mathcal{V}_N^{\text{JC}}$ its complex Zariski closure. A point is *regular* when the parameterization has its generic Jacobian rank there. We say two model images have a *full-dimensional regular overlap* when their intersection contains a relative neighborhood of a common point whose dimension equals the dimension of both images.

For the two semi-directed topologies below, $\mathcal{M}_N^{\text{JC}}$ is computed from the displayed LSA-valid rooting in (2). Root suppression combines the two root-edge multipliers into the single effective multiplier $x_{AC} = x_{\rho A}x_{\rho C}$. Conversely, every $x_{AC} \in (0, 1)$ has strict factorizations in $(0, 1)^2$ (for example, $\sqrt{x_{AC}}\sqrt{x_{AC}}$). Thus this effective parameterization has exactly the same local stochastic image as the rooted presentation; no root-boundary parameter is introduced.

3 The four-leaf pair

Let the internal vertices be a root ρ , tree vertices A, B, D, E , and reticulations C, F . Both rooted networks have the internal arcs

$$\rho \rightarrow A, \quad \rho \rightarrow C, \quad A \rightarrow B, \quad B \rightarrow C, \quad C \rightarrow D, \quad D \rightarrow E, \quad A \rightarrow F, \quad E \rightarrow F. \quad (2)$$

Their pendant arcs are

$$N : \quad B \rightarrow 1, \quad D \rightarrow 2, \quad F \rightarrow 3, \quad E \rightarrow 4, \quad (3)$$

$$N' : \quad E \rightarrow 1, \quad D \rightarrow 2, \quad F \rightarrow 3, \quad B \rightarrow 4. \quad (4)$$

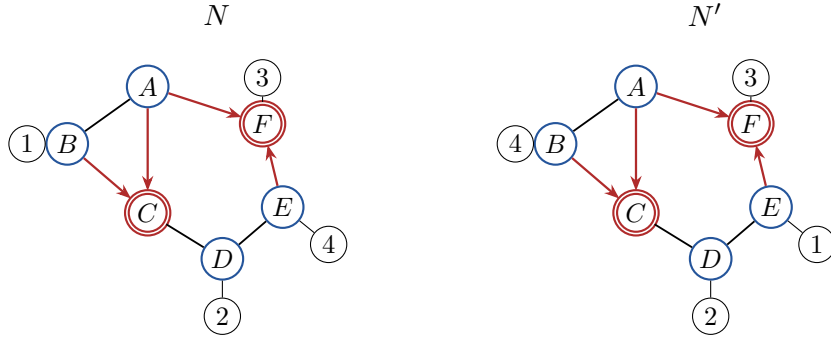


Figure 1: The standard semi-directed reductions. Double circles are reticulations, and arrowheads are retained only on incoming reticulation edges. The unique triangle is $A-B-C$.

After the root is suppressed, the nontrivial blob is the theta graph with three $A-C$ paths

$$A - C, \quad A - B - C, \quad A - F - E - D - C.$$

Its simple cycles have lengths 3, 5, 6.

In the formulas below, $x_{AC} = x_{\rho A}x_{\rho C}$ is the effective edge created by root suppression. The probability λ_C selects that $A-C$ parent edge and $1 - \lambda_C$ selects $B \rightarrow C$; similarly, λ_F selects $A \rightarrow F$ and $1 - \lambda_F$ selects $E \rightarrow F$.

Proposition 3.1 (Combinatorial certification). *The rooted presentations (2)–(4) are binary, level two, and tree-child. Their standard reductions lie in $W_{\text{TC}} \setminus S_{\text{TC}}$. The two reductions are nonisomorphic and are not $\mathbb{T}$ -equivalent.*

Proof. The bidegrees in the displayed arc lists are $(0, 2)$ at ρ , $(1, 2)$ at A, B, D, E , $(2, 1)$ at C, F , and $(1, 0)$ at the leaves. The lists are acyclic. In each rooted presentation every internal vertex has a tree-or-leaf child, so the presentation is tree-child. The unique blob has the three paths above and contains exactly the two reticulations C, F ; hence it is level two.

The displayed rooting proves weak tree-childness. To see directly that the mixed graph is not strongly tree-child, insert a new root on the retained arc $B \rightarrow C$. The internal arcs of the resulting presentation are

$$\rho' \rightarrow B, \quad \rho' \rightarrow C, \quad B \rightarrow A, \quad A \rightarrow C, \quad A \rightarrow F, \quad C \rightarrow D, \quad D \rightarrow E, \quad E \rightarrow F.$$

The pendant arcs are those in (3) or (4), respectively. This is binary and acyclic, its root is the LSA (the pendant leaf at B and the pendant leaf at D lie on opposite root branches), and applying sd_0 recovers the mixed graph. But A has the two reticulations C, F as its children and hence is not tree-child. In the terminology of Holtgreffe et al. [6], A is an omnian. Exhaustive rooting enumeration finds five admissible rootings in total, only two of which are tree-child.

In N , leaf 1 is adjacent to B , which lies on the unique triangle. In N' , leaf 1 is adjacent to E , which lies on no triangle. This leaf-labelled graph property is unchanged by isomorphism or by a redirection confined to the triangle. Thus the reductions are neither isomorphic nor $\mathbf{T}$ -equivalent. $\square$

4 The common eight-dimensional variety

The automorphism group of G permutes its three nonzero characters. The zero-sum four-leaf coordinates have fifteen JC symmetry orbits, including normalization. We denote the fourteen nonconstant orbits by

$$(A, B, C, D, E, F, G, H, J, K, L, M, N, O).$$

One explicit choice of representatives is recorded in [appendix A](#). Equality of these fourteen entries is equivalent to equality of the complete Fourier tensor.

Proposition 4.1 (Six common equations). *The complete displayed-tree Fourier parameterizations of both N and N' satisfy*

$$J - K - M + N = 0, \tag{5}$$

$$J - AH - BF + CE = 0, \tag{6}$$

$$GL - EN = 0, \tag{7}$$

$$L^2 - BEH = 0, \tag{8}$$

$$BM - DL - B^2F + BCE = 0, \tag{9}$$

$$BEO - BGH - CEL + DEH = 0. \tag{10}$$

Proof. Each network has four displayed trees. Substitute their split monomials from (1), sum with the two inheritance parameters, and expand the six left-hand sides. Every coefficient is zero. The release verifier performs this calculation from the graph encodings rather than accepting (5)–(10) as input. $\square$

Lemma 4.2 (Implicit geometry). *On $BE \neq 0$, equations (5)–(10) define an irreducible eight-dimensional variety. Its positive real locus is smooth.*

Proof. Five equations reconstruct

$$J = AH + BF - CE, \quad N = GL/E, \quad (11)$$

$$M = (DL + B^2F - BCE)/B, \quad K = J + N - M, \quad (12)$$

$$O = (BGH + CEL - DEH)/(BE). \quad (13)$$

The remaining equation is $L^2 = BEH$. The localized coordinate ring is

$$\mathbb{C}[A, B, C, D, E, F, G, H, L, (BE)^{-1}]/(L^2 - BEH).$$

The defining quadratic is irreducible over the rational function field in $A, \dots, H$, so the ring is an integral domain of dimension eight. Since $\partial(L^2 - BEH)/\partial H = -BE$, it is smooth on $BE \neq 0$. On the open stochastic locus every Fourier coordinate here is a positive sum of positive monomials, so $B, E, H, L > 0$ and the physical sheet is $L = +\sqrt{BEH}$. $\square$

5 An exact common interior point

The two root edges are represented after root suppression by the effective multiplier $x_{AC} = x_{\rho A}x_{\rho C}$. The strict rooted realization

$$x_{\rho A} = \frac{2}{3}, \quad x_{\rho C} = \frac{3}{4}$$

gives $x_{AC} = 1/2$.

For N , set

$$\begin{aligned} x_{D2} &= \frac{1}{2}, & x_{DE} &= \frac{2}{5}, & x_{E4} &= \frac{3}{8}, & x_{EF} &= \frac{1}{3}, & x_{F3} &= \frac{1}{2}, \\ x_{CD} &= \frac{9}{20}, & x_{AB} &= \frac{3}{5}, & x_{B1} &= \frac{1}{5}, \\ x_{AC} &= x_{AF} = x_{BC} = \frac{1}{2}, & \lambda_C &= \lambda_F = \frac{1}{2}. \end{aligned} \quad (14)$$

Let β be the smaller root of

$$43337075\beta^2 - 36083110\beta + 7336259 = 0, \quad \frac{441}{1250} < \beta < \frac{3529}{10000}. \quad (15)$$

For N' , set

$$\begin{aligned} x_{D2} &= \frac{1}{2}, & x_{DE} &= \frac{171}{775}, & x_{AF} &= \frac{10339}{53010\beta}, \\ x_{AB} &= \frac{24835\beta}{20678 - 24835\beta}, & x_{F3} &= \frac{1767}{4832}, & x_{CD} &= \frac{9934}{12215}, \\ x_{E1} &= \frac{31}{190}, & x_{B4} &= \frac{3}{20\beta}, \\ x_{AC} &= x_{EF} = x_{BC} = \frac{1}{2}, & \lambda_C &= \lambda_F = \frac{1}{2}. \end{aligned} \quad (16)$$

The polynomial in (15) has opposite signs at the rational endpoints, and the interval lies strictly below its axis of symmetry. It therefore isolates the smaller root. Exact rational interval arithmetic then proves that every multiplier in (16) lies strictly in $(0, 1)$; no boundary or merely entrywise-positive transition matrix is used.

At these two parameter points the common orbit vector is

$$(A, \dots, H) = \left(\frac{277}{8000}, \frac{3}{40}, \frac{67}{2400}, \frac{127}{16000}, \frac{27}{5000}, \frac{81}{5000}, \frac{153}{160000}, \frac{9}{500} \right), \quad (17)$$

$$(J, \dots, O) = \left(\frac{27}{16000}, \frac{261}{320000}, \frac{27}{10000}, \frac{27}{20000}, \frac{153}{320000}, \frac{183}{80000} \right). \quad (18)$$

To certify regularity, fix

$$x_{AC} = x_{AF} = x_{BC} = \lambda_C = \lambda_F = \frac{1}{2}$$

for N and use

$$(P, s, Q, t, R, u, v, S) = (x_{D2}, x_{DE}, x_{EA}, x_{EF}, x_{F3}, x_{CD}, x_{AB}, x_{B1}).$$

The Jacobian determinant of $(A, \dots, H)$ is

$$-\frac{P^3 s^3 Q^4 t R^4 u^3 v S^3 (s-1)^2 (v-1)(v+1)^2}{16384}, \quad (19)$$

which is nonzero throughout the open gauge cube.

For N' , fix

$$x_{AC} = x_{EF} = x_{BC} = \lambda_C = \lambda_F = \frac{1}{2}$$

and use

$$(P', x, y, z, R', w, S', Q') = (x_{D2}, x_{DE}, x_{AF}, x_{AB}, x_{F3}, x_{CD}, x_{E1}, x_{B4}).$$

The determinant is

$$-\frac{(P')^3 x^2 y^2 z (R')^4 w^4 (S')^3 (Q')^4 (x-1)^2 (z-1)(z+1)^3}{32768}, \quad (20)$$

which is also nonzero on the open cube.

Proposition 5.1 (Four-leaf stochastic overlap). *The complete JC Fourier tensors of N at (14) and N' at (16) agree exactly. The two stochastic model images share a regular relatively open eight-dimensional neighborhood.*

Proof. Direct displayed-tree evaluation in $\mathbb{Q}(\beta)$ gives (17)–(18) for both networks. The exact verifier checks all 256 Fourier entries, including the forced zero coordinates; invertibility of the Fourier transform gives equality of all pattern probabilities.

By [proposition 4.1](#) and [lemma 4.2](#), each model lies in the same irreducible eight-dimensional variety. The nonzero determinants (19) and (20) show that each image is an eight-dimensional local submanifold at the common point. The Zariski closure of each parameterized image is irreducible; containment in an irreducible eight-dimensional locus together with rank eight therefore makes both closures equal to that locus. Moreover, the first eight coordinates of each stochastic image contain a neighborhood of the vector in (17). Intersect the two neighborhoods. Positivity forces the same value $L = +\sqrt{BEH}$, and (13) forces the other five coordinates. Hence the intersection contains a relative open eight-dimensional neighborhood. $\square$

Corollary 5.2. *The set of parameters of N admitting a realization on N' contains a nonempty Euclidean-open subset of the full parameter space. It is not contained in any proper algebraic subset.*

Proof. The full parameterization has rank eight at the source point because the restricted gauge map has rank eight. The submersion theorem maps a neighborhood of the source point onto a relative neighborhood of the common variety. The inverse image of the open overlap in [proposition 5.1](#) is therefore nonempty and open. A nonzero real polynomial cannot vanish on a Euclidean-open parameter set. $\square$

6 Leaf substitution

Let P be a normalized positive Fourier tensor on $X \cup \{a\}$. Replace the leaf a by a cherry with leaves a_1, a_2 and new pendant multipliers $u, v \in (0, 1)$. The former pendant edge becomes the stem of the cherry. The new tensor is

$$\tilde{P}(g_X, g_1, g_2) = P(g_X, g_1 \oplus g_2) u^{\mathbf{1}_{[g_1 \neq 0]}} v^{\mathbf{1}_{[g_2 \neq 0]}}. \quad (21)$$

Lemma 6.1 (Positive analytic inverse). *The map $(P, u, v) \mapsto \tilde{P}$ in (21) is a positive real-analytic embedding. Its inverse is real analytic on its image. Consequently, substituting a cherry adds exactly two model dimensions and preserves full-dimensional regular overlap.*

Proof. Fix $h \in G \setminus \{0\}$. With all characters on X equal to zero,

$$uv = \tilde{P}(0, h, h),$$

because $P(0, 0) = 1$. Choose any g_X with total character h . Positivity of the open JC Fourier tensor permits division, and

$$\frac{u}{v} = \frac{\tilde{P}(g_X, h, 0)}{\tilde{P}(g_X, 0, h)}.$$

Thus u and v are recovered uniquely by the positive square root. Every entry of P is then recovered from (21) by division by a positive monomial in u, v . These formulas are real analytic. Therefore the image is analytically isomorphic to the product of the base tensor image with $(0, 1)^2$, proving the dimension and regularity assertions. $\square$

Proof of theorem 1.1. The case $n = 4$ is [propositions 3.1](#) and [5.1](#). For each new taxon, replace the same labelled leaf in both networks by an identically labelled cherry. Equation (21) carries every common base distribution and every common pair (u, v) to the same enlarged distribution. By [lemma 6.1](#), a common regular open region gains exactly two dimensions at each step. Starting from dimension eight gives

$$8 + 2(n - 4) = 2n.$$

The displayed tree-child rootings extend through each inserted cherry, so the enlarged semi-directed topologies remain in W_{TC} . The original omnian at A is unchanged, so they remain outside S_{TC} . The substituted leaf block attaches at a triangle vertex in one topology and at a nontriangle vertex in the other. This labelled cut-component property is preserved by mixed-graph isomorphism and by T . Thus the two enlarged topologies remain nonisomorphic and non- T -equivalent. Repeating the construction proves the theorem for every $n \geq 4$. $\square$

7 Scope and biological interpretation

The theorem separates three notions that can otherwise be conflated. The two models have equal complex closures, they have a full-dimensional regular stochastic overlap, and the exhibited parameter correspondence is strictly inside Θ_0 . We do not claim equality of the complete stochastic images, nor that the overlap is dense in either image.

The ambiguity is also not root relocation or triangle orientation. It changes which labelled descendant component is attached to the triangle. Consequently, even an exact infinite-data JC distribution cannot always determine that attachment in the weakly tree-child level-2 class. The construction does not settle identifiability inside S_{TC} ; that positive classification remains outside the scope of this paper.

8 Exact verification

The accompanying reproducibility package contains two implementations. The primary symbolic verifier derives displayed-tree Fourier parameterizations from machine-readable rooted arc lists, checks the six identities, verifies the two Jacobian factorizations, isolates β , proves strict positivity, and checks all 256 coordinates. The independent verifier shares no graph, Fourier, isomorphism, or number-field code with the primary implementation. It additionally enumerates all admissible rootings and verifies the all- n substitution argument at the level of exact formulas.

Every released certificate is regenerated from source. Hashes and exact commands are recorded in the package manifest. Finite graph regressions for several values of n are included only as error-detection tests; the quantified all- n result is the proof of [lemma 6.1](#) and induction, not finite enumeration.

A Fourier orbit convention and algebraic data

Represent the nonzero elements of G by 1, 2, 3. The fourteen nonconstant JC orbits can be represented by

symbol	tuple	name	symbol	tuple	name
A	0011	r_{34}	B	0101	r_{24}
C	0110	r_{23}	D	0123	t_{234}
E	1001	r_{14}	F	1010	r_{13}
G	1023	t_{134}	H	1100	r_{12}
J	1111	u_{1111}	K	1122	u_{1122}
L	1203	t_{124}	M	1212	u_{1212}
N	1221	u_{1221}	O	1230	t_{123}

The field is $\mathbb{Q}(\beta)$ with the minimal polynomial in [\(15\)](#). Reduction uses

$$\beta^2 = \frac{36083110}{43337075}\beta - \frac{7336259}{43337075}.$$

The independent verifier represents every field element uniquely as $a + b\beta$ with $a, b \in \mathbb{Q}$. Equality is coefficient equality after this reduction. Order is never inferred from the algebraic representation; all stochastic inequalities are proved separately using the rational isolating interval in [\(15\)](#).

Data and code availability

The exact graph encodings, verifiers, generated certificates, source, and deterministic reproduction command accompany this manuscript in the project repository. No specialized phylogenetic-classification software is needed.

Use of generative AI

Generative AI systems assisted with exploratory algebra, implementation, drafting, and adversarial review. Earlier automated drafts overclaimed a global positive classification and used an incorrect bridge-coordinate argument; those claims were withdrawn. The present manuscript contains only the theorem supported by the exact, independently replayed certificates. The author is responsible for the statement, proof, code, and final submission.

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